HU Collection · Kevinvariant Pathways

Lengths and ratios

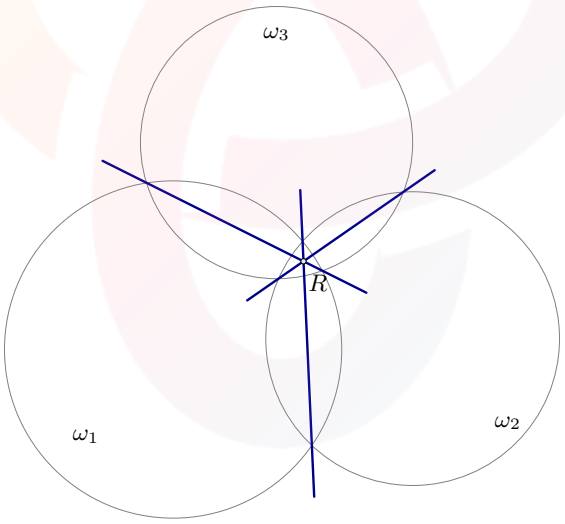
Level 4 Olympiad · Geometry

Power of a point, radical axes, the ratio lemma with Ceva and Menelaus, and mass points, for AIME computation and BMO proofs.

By the end you can: You will be able to use signed power of a point, radical axes and radical centres in proofs and computations; You will be able to turn angle data into side ratios with the ratio lemma and trig Ceva; You will be able to read off ratios and areas in cevian configurations with mass points.

Lesson

Three circles meet in pairs, and you draw the three common chords. They always pass through one point. Try to prove it by angle chasing and you will get nowhere: there is no useful angle to chase. The proof is one line, once you have the right quantity, the *power of a point*. This unit collects the tools that turn geometry into lengths and ratios: power of a point, radical axes, the ratio lemma with Ceva and Menelaus, and mass points. They are the workhorses of AIME geometry and of every BMO1 problem that asks about lengths.



Power of a point

Theorem. Let ω have centre O and radius r, and let P be any point. If a line through P meets ω at A and B, then

$$\overline{PA} \cdot \overline{PB} = OP^2 - r^2,$$

where bars denote signed lengths along the line, so the product is negative exactly when P is inside the circle. The number $\mathrm{pow}(P) = OP^2 - r^2$ is the *power* of P. If PT is a tangent, $PT^2 = \mathrm{pow}(P)$.

Proof: let M be the foot of the perpendicular from O to the line, which is the midpoint of AB. With signed lengths, $\overline{PA} \cdot \overline{PB} = (\overline{PM} + \overline{MA})(\overline{PM} - \overline{MA}) = PM^2 - MA^2 = (OP^2 - OM^2) - (OA^2 - OM^2) = OP^2 - r^2$. The tangent is the case $A = B = T$.

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The converse matters as much. If lines AB and CD meet at P and $\overline{PA}\cdot\overline{PB}=\overline{PC}\cdot\overline{PD}$, then A,B,C,D are concyclic: line PC meets circle ABC again at a point D' with $\overline{PC}\cdot\overline{PD'}=\overline{PA}\cdot\overline{PB}$, so $D'=D$. Likewise, $PT^2=\overline{PA}\cdot\overline{PB}$ makes PT tangent to circle ABT.

Radical axis and radical centre

Radical axis. For two circles with different centres, the points with equal power with respect to both form a line perpendicular to the line of centres. If the circles meet at X and Y, it is line XY; if they touch, it is the common tangent at the point of contact.
Radical centre. For three circles with non-collinear centres, the three radical axes meet at one point.

Put the centres at $(0,0)$ and $(d,0)$. The condition $x^2+y^2-r_1^2=(x-d)^2+y^2-r_2^2$ simplifies to $x=\dfrac{d^2+r_1^2-r_2^2}{2d}$, a line perpendicular to the line of centres, and common points of the circles (power 0 for both) lie on it. For three circles, two of the axes are perpendicular to non-parallel lines, so they meet at a point R; then R has the same power for all three circles, so it lies on the third axis. That is the opening problem.

In practice: a point on the common chord line has equal tangents to both circles; four points are concyclic when two lines through them meet at a point with equal products; three lines are concurrent when they are radical axes of three circles.

The ratio lemma, Ceva and Menelaus

Ratio lemma. If D lies on side BC of triangle ABC, then

$$\frac{BD}{DC}=\frac{AB\,\sin\angle BAD}{AC\,\sin\angle DAC}.$$

Ceva. For D,E,F on sides BC,CA,AB, the lines AD,BE,CF are concurrent if and only if $\dfrac{BD}{DC}\cdot\dfrac{CE}{EA}\cdot\dfrac{AF}{FB}=1$. In trigonometric form, if and only if $\dfrac{\sin\angle BAD}{\sin\angle DAC}\cdot\dfrac{\sin\angle CBE}{\sin\angle EBA}\cdot\dfrac{\sin\angle ACF}{\sin\angle FCB}=1$.
Menelaus. If a line meets lines BC,CA,AB at D,E,F, the same product of unsigned ratios equals 1; conversely, if the product is 1 and one or three of the points lie outside the sides, the points are collinear.

The ratio lemma is an area comparison: triangles ABD and ADC share the height from A, so $\dfrac{BD}{DC}=\dfrac{[ABD]}{[ADC]}=\dfrac{\frac12 AB\cdot AD\sin\angle BAD}{\frac12 AC\cdot AD\sin\angle DAC}$. The angle bisector theorem is the special case of equal angles. For Ceva, if the cevians meet at P then $BD/DC=[ABP]/[ACP]$ (subtract the two pairs of triangles with bases BD, DC), and the three such ratios multiply to 1; the converse follows because the ratio determines the point. Trig Ceva is Ceva with the ratio lemma applied three times: the side lengths cancel. Use the ratio lemma whenever you know angles at a vertex and want a ratio on the opposite side, or the reverse.

Mass points

Mass points are bookkeeping for Ceva configurations. Put masses $m_A,m_B,m_C>0$ at the vertices. The centre of mass of B and C is the point D of BC with $BD:DC=m_C:m_B$ (the lever law: the heavier mass is nearer), and it carries mass m_B+m_C. The centre of mass $P=\dfrac{m_A A+m_B B+m_C C}{m_A+m_B+m_C}$ of all three is also the centre of mass of A and D, so P lies on AD with $AP:PD=(m_B+m_C):m_A$.

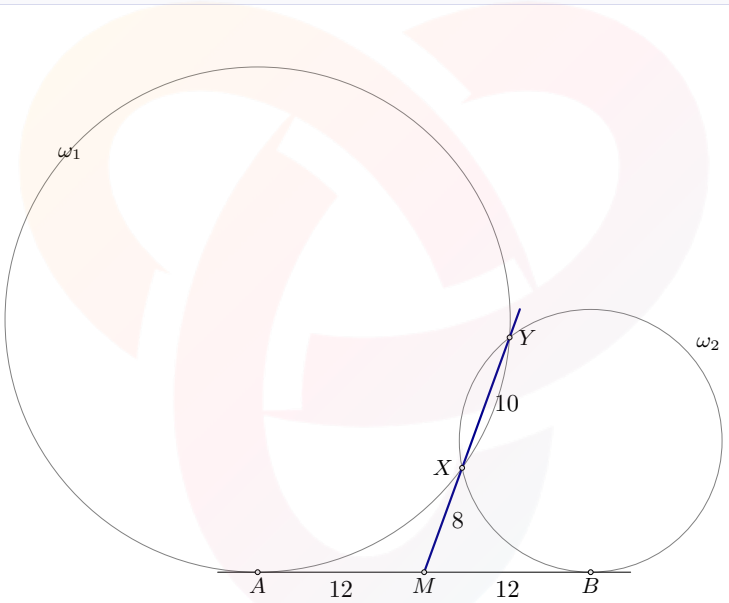
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Grouping the masses differently shows P is on the other two cevians as well. Moreover $[PBC] : [PCA] : [PAB] = m_A : m_B : m_C$. So: choose masses to match two given ratios, and every other ratio and area in the figure can be read off.

Worked examples

Example 1. Circles ω_1 and ω_2 meet at X and Y. A common tangent, nearer to X than to Y, touches ω_1 at A and ω_2 at B. Given $AB = 24$ and $XY = 10$, find the distance from X to the midpoint of AB.

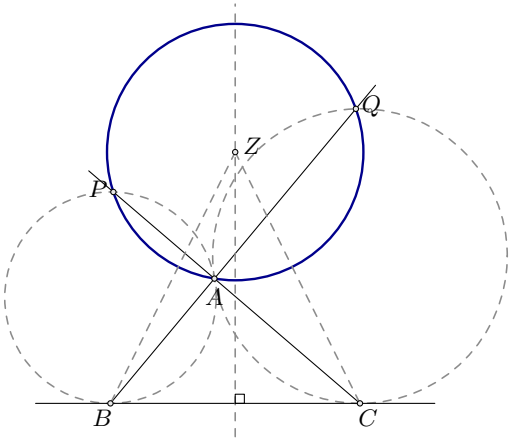
Solution.
Coordinates look hopeless: we know neither radius. The only link between the circles is the common chord, so think about powers. Let line XY meet AB at M. Since M is on the radical axis, $MA^2 = MX \cdot MY = MB^2$, so M is the midpoint of AB and $MA = 12$.
The tangent line lies outside both circles, so M has positive power and X, Y are on the same side of M, with X nearer. Writing $MX = x$ gives $x(x + 10) = 144$, so $(x - 8)(x + 18) = 0$ and $x = 8$.



Example 2. In triangle ABC, the circle through A tangent to BC at B meets line AC again at P, and the circle through A tangent to BC at C meets line AB again at Q. Prove that the circumcentre of triangle APQ is equidistant from B and C.

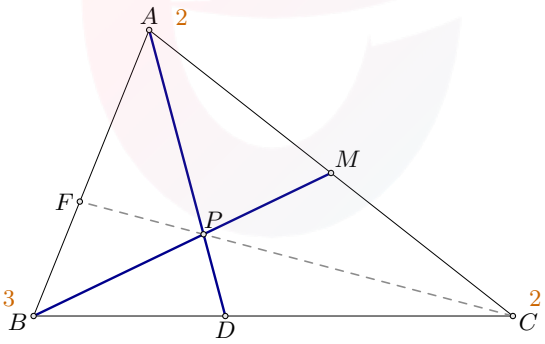
Solution.
You met this configuration in L4-GE1, where angles produced a fixed point. Angle chasing here finds many equal angles but nothing about the circumcentre Z. Rephrase instead: if ρ is the radius of circle APQ, then $ZB^2 - \rho^2$ and $ZC^2 - \rho^2$ are the powers of B and C. So it is enough to show that B and C have equal power with respect to circle APQ.
B lies on line AQ, so its power is $\overline{BQ} \cdot \overline{BA}$. But Q and A also lie on the circle tangent to BC at C, and BC is a tangent from B to that circle, so $\overline{BQ} \cdot \overline{BA} = BC^2$. In the same way the power of C is $\overline{CP} \cdot \overline{CA} = CB^2$. The powers are equal, so $ZB = ZC$. Signed lengths mean no case analysis: wherever P and Q fall on their lines, the products are the powers.

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Example 3. In triangle ABC of area 21, $AB = 6$ and $AC = 9$. The bisector of angle A meets BC at D, the median from B meets AC at its midpoint M, and AD meets BM at P. Line CP meets AB at F. Find $AP : PD$, $BP : PM$, $AF : FB$ and the area of triangle APB.

Solution.
Coordinates would work but are slow, and BC is not even given. The data are two ratios on two sides, which is exactly the mass point situation. By the angle bisector theorem $BD : DC = 6 : 9 = 2 : 3$, so we need $m_C : m_B = 2 : 3$; take $m_B = 3$, $m_C = 2$. As M is the midpoint of AC, $m_A = m_C = 2$.
Now read everything off: $AP : PD = (m_B + m_C) : m_A = 5 : 2$, $BP : PM = (m_A + m_C) : m_B = 4 : 3$ and $AF : FB = m_B : m_A = 3 : 2$. For the area, $[APB] : [ABC] = m_C : (m_A + m_B + m_C) = 2 : 7$, so $[APB] = 6$. Check with Ceva: $\frac{2}{3} \cdot 1 \cdot \frac{3}{2} = 1$. The classic slip is to put $BD : DC = m_B : m_C$; that is backwards.



Example 4. Triangle ABC has $AB = 6$, $AC = 8$, $BC = 7$. The tangent to its circumcircle at A meets line BC at T. (a) Find TA. (b) Prove that every point X on the circle Γ with centre T through A satisfies $XB : XC = 3 : 4$.

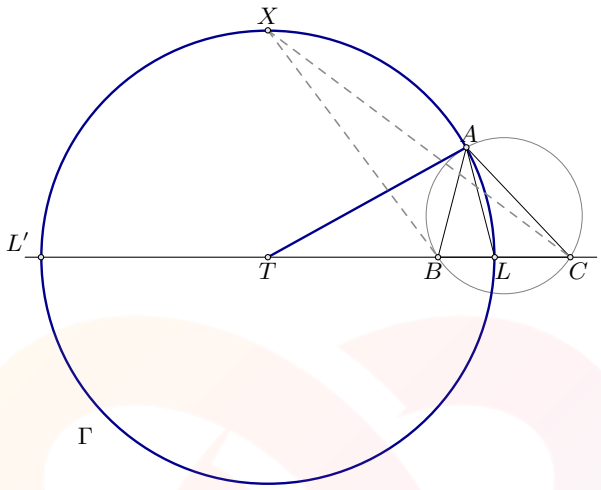
Solution.
(a) One could find angles with the cosine rule, but the tangent hands us a similarity. Since $AB < AC$ we have $\angle C < \angle B$, and the tangent at A meets line BC beyond B. By the alternate segment theorem $\angle TAB = \angle ACB$, and triangles TAB, TCA share the angle at T, so they are similar and

$$\frac{TA}{TC} = \frac{TB}{TA} = \frac{AB}{CA} = \frac{3}{4}.$$

Thus $TB = \frac{9}{16}TC$ and $TC - TB = 7$. This gives $TC = 16$, $TB = 9$, and $TA = 12$; indeed $TA^2 = 144 = TB \cdot TC$.

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(b) For X on Γ but not on line BC, $TX^2 = TA^2 = TB \cdot TC$, so by the converse of power of a point TX is tangent to circle XBC. Exactly as in (a), $\triangle TXB \sim \triangle TCX$, so $XB : XC = TX : TC = 12 : 16 = 3 : 4$. The two points of Γ on line BC are L with $BL = 3$, $LC = 4$, and L' with $L'B = 21$, $L'C = 28$, which also fit.
So Γ is the Apollonius circle of the ratio $3 : 4$. Note that L is the foot of the bisector from A, so $TA = TL$: worth remembering for AIME.



Common traps.

- Forgetting signs: inside the circle, $PA \cdot PB = r^2 - OP^2$. Signed lengths handle every case at once.
- Putting mass points backwards: $BD : DC = m_C : m_B$.
- Applying Ceva to points on extensions, or Menelaus to points all on the sides, without care.
- Ratio lemma with the wrong angles: they are the angles at A, not at D.
- In AIME computations, keeping the wrong root of a quadratic or the wrong configuration (is T beyond B or beyond C?).

How to spot this in a contest

Products of lengths along lines through one point mean power of a point. Two circles with a common chord or a common tangent suggest the radical axis; three circles, or three lines that should be concurrent, suggest the radical centre. A tangent at a vertex meeting the opposite side gives the similarity $TAB \sim TCA$. Angles at a vertex together with a ratio on the opposite side call for the ratio lemma or trig Ceva. Cevians with given ratios, or areas cut by cevians, are mass points. Finally, compute in exact fractions and check the answer against a rough scale drawing.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ Circles ω_1 and ω_2 have radii 13 and 15, and their centres are 14 apart. A point P on the line

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through their two common points is at distance 20 from the line of centres. Find the length of a tangent from P to ω_1.

2. ⋆⋆ Triangle ABC has area 120. Point D lies on BC with $BD : DC = 1 : 2$, and E lies on CA with $CE : EA = 1 : 3$. The segments AD and BE meet at P. Find the area of quadrilateral $PDCE$.

3. ⋆⋆ In triangle ABC, $AB = 4$, $AC = 6$ and $\angle BAC = 120^\circ$. The point D on BC satisfies $\angle BAD = 90^\circ$. Find BD/DC.

4. ⋆⋆⋆ From a point P outside a circle ω with centre O, the tangents touch ω at A and B, and M is the midpoint of AB. A line through P, not through O, meets ω at C and D. Prove that O, M, C, D are concyclic.

5. ⋆⋆⋆ Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$. The circle with diameter BC meets AB again at F and AC again at E. Line EF meets line BC at T. Find TB.

6. ⋆⋆⋆⋆ Let ABC be an acute triangle with $AB \neq AC$, orthocentre H, and let M be the midpoint of BC. The circle through A tangent to BC at B and the circle through A tangent to BC at C meet again at K. Prove that K lies on AM and that $HK \perp AM$.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AIME 1985 Problem 6 ⋆⋆
2. BMO1 2009/10 Problem 4 ⋆⋆
3. AIME 1989 Problem 15 ⋆⋆⋆
4. BMO1 2007/08 Problem 5 ⋆⋆⋆
5. BMO1 2012/13 Problem 6 ⋆⋆⋆
6. AIME II 2016 Problem 10 ⋆⋆⋆
7. USAJMO 2012 Problem 1 ⋆⋆⋆
8. BMO1 2025/26 Problem 4 ⋆⋆⋆
9. AIME I 2019 Problem 13 ⋆⋆⋆
10. BMO1 2016/17 Problem 5 ⋆⋆⋆
11. AIME I 2016 Problem 15 ⋆⋆⋆⋆
12. AIME I 2019 Problem 15 ⋆⋆⋆⋆

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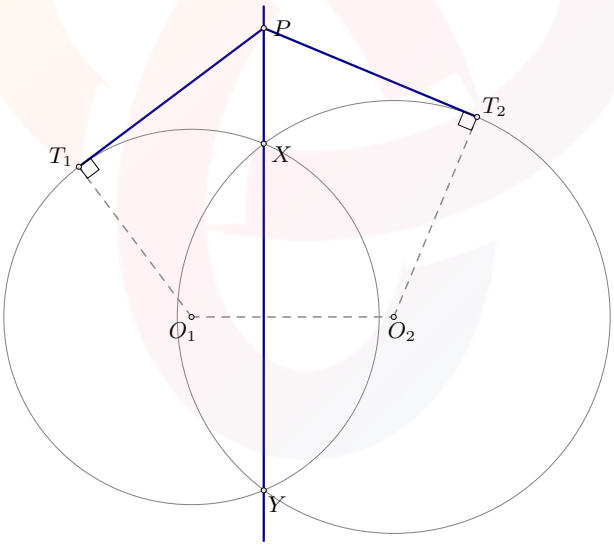
Hints

1. Locate the radical axis with the formula, then the power of P is $O_1P^2 - 13^2$.
2. Choose masses to match both ratios; the masses are proportional to the areas $[PBC]$, $[PCA]$, $[PAB]$.
3. You know both angles at A on either side of AD.
4. Express PA^2 in two ways: once with the circle, once with the right-angled triangle PAO.
5. Menelaus works. For a slicker route, line EF is the radical axis of the circle on diameter BC and another circle through E and F that meets BC at known points.
6. For the first part, compare the powers of M. For the second, show K is on circle BHC, and find a diameter of that circle with one end on line AM.

Solutions

Problem 1

Answer: 16



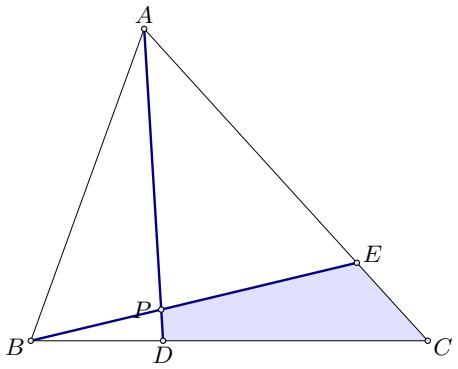
Put $O_1 = (0, 0)$ and $O_2 = (14, 0)$. The radical axis is the line $x = \dfrac{14^2 + 13^2 - 15^2}{2 \cdot 14} = 5$, so $P = (5, 20)$ (or its mirror image, which gives the same answer). The square of the tangent length is the power $O_1P^2 - 13^2 = 25 + 400 - 169 = 256$, so the tangent has length 16.

As a check, the power with respect to ω_2 is $9^2 + 20^2 - 15^2 = 256$ too, as it must be for a point on the radical axis.

Problem 2

Answer: 26

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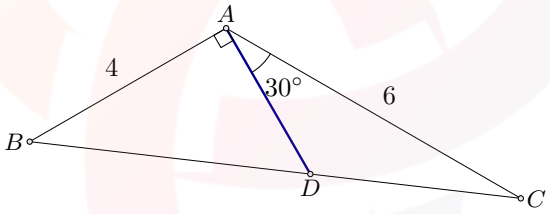


We need $m_C : m_B = BD : DC = 1 : 2$ and $m_A : m_C = CE : EA = 1 : 3$. Take $m_A = 1$, $m_C = 3$, $m_B = 6$, with total 10. Then P is the centre of mass and $[PBC] : [PCA] : [PAB] = 1 : 6 : 3$, so $[PBC] = 12$ and $[PCA] = 72$.

Triangles PDC and PBC share the height from P, so $[PDC] = \frac{DC}{BC}[PBC] = \frac{2}{3} \cdot 12 = 8$. Similarly $[PCE] = \frac{CE}{CA}[PCA] = \frac{1}{4} \cdot 72 = 18$. Hence $[PDCE] = 8 + 18 = 26$.

Problem 3

Answer: $\frac{4}{3}$



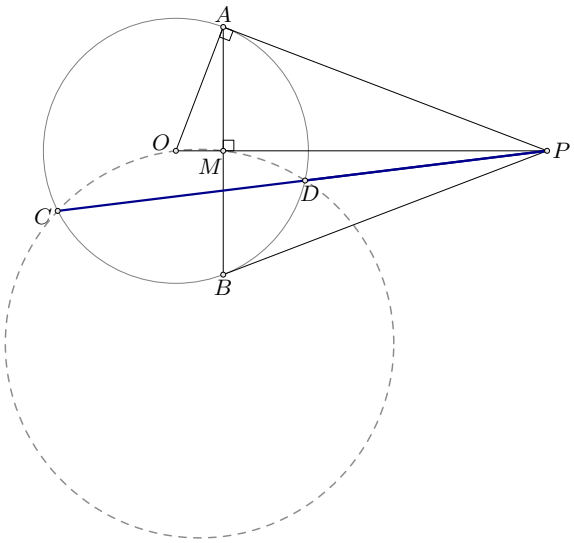
Since $\angle BAD = 90°$, $\angle DAC = 120° - 90° = 30°$. By the ratio lemma,

$$\frac{BD}{DC} = \frac{AB \sin \angle BAD}{AC \sin \angle DAC} = \frac{4 \cdot 1}{6 \cdot \frac{1}{2}} = \frac{4}{3}.$$

No side length BC was needed: that is the point of the ratio lemma.

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Problem 4

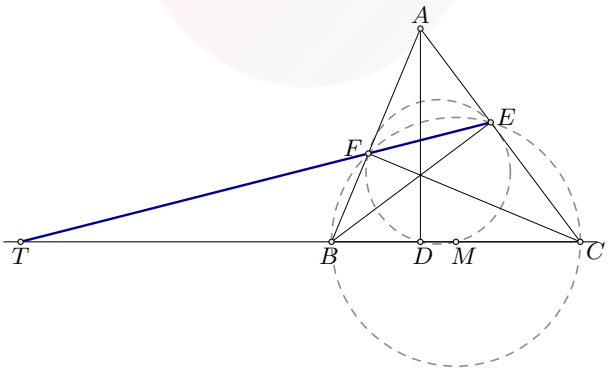


Since $PA = PB$ and $OA = OB$, both P and O lie on the perpendicular bisector of AB. So line PO is that bisector, it passes through the midpoint M, and $AM \perp PO$. The radius OA is perpendicular to the tangent PA, so triangle PAO has a right angle at A, and AM is its altitude. Triangles PMA and PAO are similar (they share the angle at P and each has a right angle), giving $PM \cdot PO = PA^2$.

By power of a point, $PA^2 = PC \cdot PD$. Therefore $PM \cdot PO = PC \cdot PD$. Here M and O lie on one ray from P, and C and D lie on another ray from P, since P is outside the circle, so the equality holds for signed lengths as well. The two lines are different because the line PCD does not pass through O. By the converse of power of a point, O, M, C, D are concyclic. ■

Problem 5

Answer: $\frac{35}{2}$



The angles $\angle BFC$ and $\angle BEC$ are right, so F and E are the feet of the altitudes from C and B. Let D be the foot of the altitude from A and M the midpoint of BC; the familiar 13-14-15 split gives $BD = 5$, $DC = 9$, and $BM = 7$.

The nine-point circle passes through D, E, F, M, and the circle with diameter BC passes through B, C, E, F. Their radical axis is line EF, so T has equal powers: $\overline{TB} \cdot \overline{TC} = \overline{TD} \cdot \overline{TM}$. Since E and F lie on the sides, Menelaus says T is outside segment BC, and T is on the side of B (checked below). With $TB = x$,

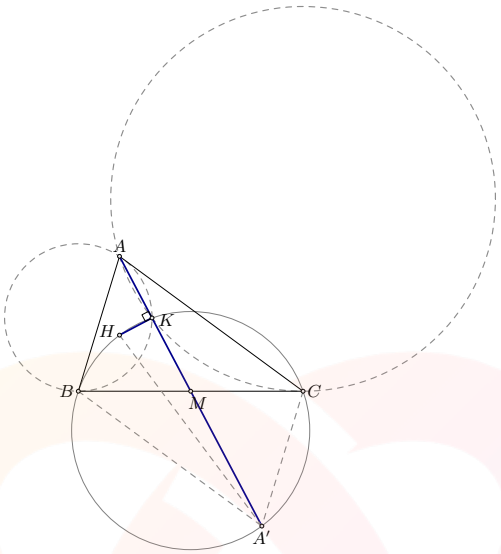
$$x(x + 14) = (x + 5)(x + 7),$$

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so $14x = 12x + 35$ and $x = \frac{35}{2}$.

Check by Menelaus: $BF = 14\cos B = \frac{70}{13}$, $AF = \frac{99}{13}$, $CE = 14\cos C = \frac{42}{5}$, $AE = \frac{33}{5}$, and $\frac{AF}{FB} \cdot \frac{BT}{TC} \cdot \frac{CE}{EA} = 1$ gives $\frac{BT}{TC} = \frac{70}{99} \cdot \frac{33}{42} = \frac{5}{9}$. So $TB < TC$, T is beyond B, and $TC - TB = 14$ gives $TB = \frac{35}{2}$ again.

Problem 6



Call the circles Γ_B (tangent at B) and Γ_C (tangent at C).

K lies on AM. The power of M with respect to Γ_B is MB^2, since MB is a tangent, and with respect to Γ_C it is MC^2. As $MB = MC$, M is on the radical axis of Γ_B and Γ_C. So is A, a common point, and $A \neq M$, so the radical axis is line AM. The common point K also lies on it, so A, K, M are collinear. Moreover $\overline{MK} \cdot \overline{MA} = MB^2$. Since $\angle BAC$ is acute, A lies outside the circle with diameter BC, so $MA > MB$; hence $0 < MK < MA$ and K lies on segment MA, with $K \neq A$.

K lies on circle BHC. Use directed angles modulo 180°. The tangent criterion in Γ_B gives $\measuredangle BKA = \measuredangle(BC, BA)$, and in Γ_C gives $\measuredangle CKA = \measuredangle(CB, CA)$. Hence

$$\measuredangle BKC = \measuredangle BKA - \measuredangle CKA = \measuredangle(BC, BA) + \measuredangle(CA, BC) = \measuredangle(CA, BA) = \measuredangle CAB.$$

Since $HB \perp AC$ and $HC \perp AB$, turning both lines through 90° gives $\measuredangle BHC = \measuredangle(AC, AB) = \measuredangle CAB$. So $\measuredangle BKC = \measuredangle BHC$ and B, K, H, C are concyclic.

The right angle. Let A' be the reflection of A in M, so $ABA'C$ is a parallelogram. Then $BA' \parallel AC \perp BH$, so $\angle HBA' = 90°$, and likewise $\angle HCA' = 90°$. Hence A' lies on circle BHC and HA' is a diameter. Now $K \neq A'$: otherwise $MK = MA$ and $MK \cdot MA = MB^2$ would give $MA = MB$, so $\angle BAC = 90°$, which is false. Also $K \neq H$, since H on the median AM would make the altitude and the median from A coincide, forcing $AB = AC$. So K is a point of the circle with diameter HA' other than its ends, and $\angle HKA' = 90°$. As K and A' lie on line AM, $HK \perp AM$. ■

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